

# Gradient Descent

Repeatedly do

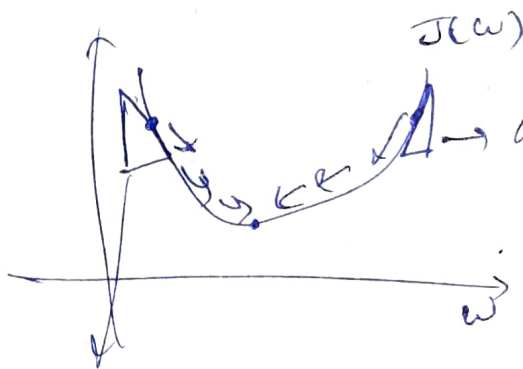
$$w := w - \alpha \frac{dT(w)}{dw}$$

learning rate  $\rightarrow$  like a hyperparameter

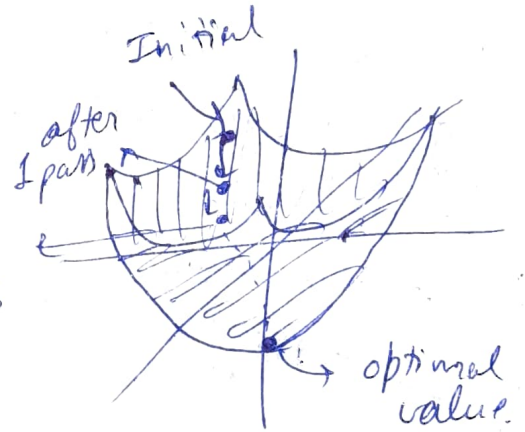
$\rightarrow$  we use  $dw$  to ~~de~~ represent this

$$w := w - \alpha dw$$

What happens is, we take an initial value and using the loss function, we optimize the value, till we get the global minimum/maxima



after 2 pass  
slope is true  
so  $w - \alpha dw$



$\rightarrow$  we slope  $\rightarrow w - \alpha$  (true number)  
 $\rightarrow$  true # of  $\alpha$  increases

Here,  $J(w)$  is used, for actual loss function, we have  $w$  &  $b$  then:

$$w := w - \alpha \frac{\partial J(w, b)}{\partial w}$$

$$b := b - \alpha \frac{\partial J(w, b)}{\partial b}$$